Uses of materials

Materials	Uses
Steel	- Construction of Bridges/ Building
Wrought/Cast iron	- Manufacturing of Tools
Gold, Silver, Copper	- Making Coins
Plastic, Wood	- Making Chairs

Why copper suitable for wiring? Copper is a good conductor of electricity, strong and ductile.

Why cooker is made up of metals? Metals are good conductor of heat, strong and does not melt with high temperature.

Hence a detailed study of properties of materials provide a steady knowledge to select things depending on their uses and nature.

9.2 ENGINEERING MATERIALS

The substances which are useful in the field of engineering are called Engineering Materials. A particular material is selected on the basis of following considerations.

1. Availability of Material
2. Cost of Material
3. Sustainability
4. Ease of Manufacture
5. Compatibility
6. Reliability
7. Recyclability

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9.2.1 Classification of Engineering Materials

Metals and Non- metals plays an important role in the engineering industry. The materials mainly used in practice are metals which may be divided into ferrous, non-ferrous and alloy. Non- metals are classified as polymer and ceramics.

9.2.2 Ferrous Metal

- The metal which contain iron as its main constituent are Ferrous metal.
- Cast iron, wrought iron and steel are some examples of ferrous metals.
- Hematite, magnetite, limonite and siderite are basically iron ores. The main ore is hematite.

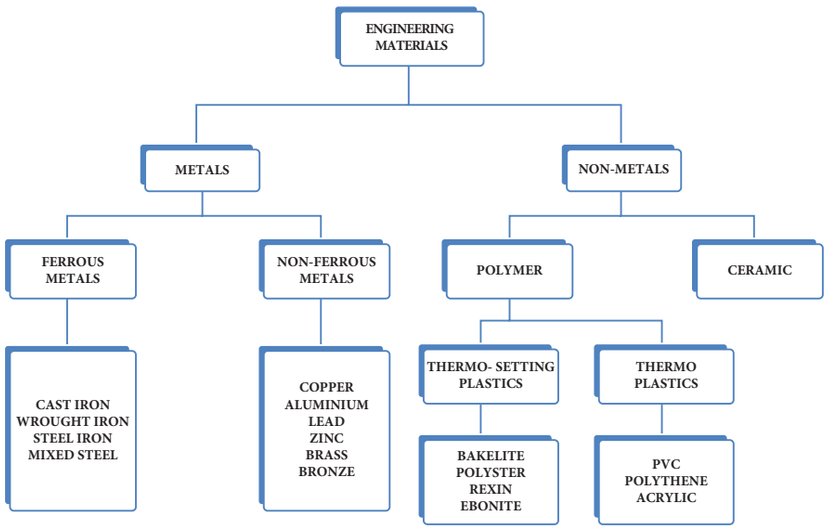


Fig 9.2 *Classification of Materials*

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